MATRIX | INDUSTRIAL MAINTENANCE

Industrial Maintenance of Closed Loop Systems - Answers

MATRIX

CP0539

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INTRODUCTION

This document highlights all the answers to the questions posed in the student handout for the worksheets curriculum “CP0539 - Industrial Maintenance of Closed Loop Systems”.

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ANSWERS

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Answers

Worksheet 1 – Closed-Loop Control Systems

Questions:

1. What is the purpose of a closed-loop flow control system?
2. What must you check before starting the system?
3. What happens when you restrict the flow using the hand valve?
4. What role does the PLC play in the system?
5. Why should you avoid making rapid changes to the valve?

Answers:

1. To continuously adjust flow using sensor feedback to meet a setpoint, maintaining accuracy and stability even if conditions change.
2. Ensure the hand valve is in the open position, with the handle in line with the piping.
3. The flow drops, the sensor detects this and sends data to the PLC, which increases pump speed and adjusts the valve to restore flow.
4. The PLC is a computer that makes real-time decisions based on sensor data to control devices like pumps and valves.
5. Rapid changes can cause oscillations in flow, making the system unstable; gradual adjustments help maintain smooth control.

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Answers

Worksheet 2 – Emergency Stops

Questions:

1. What is the main purpose of an Emergency Stop (E-Stop)?
2. How do you reset the E-Stop after it has been pressed?
3. What makes an E-Stop more reliable than a normal stop button?
4. When should an E-Stop be used?
5. Why is it important to regularly inspect the E-Stop?

Answers:

1. To quickly stop a machine or system in an emergency to protect people and equipment.
2. Twist the E-Stop anti-clockwise, then press the blue reset button or the HMI reset button.
3. It has two contact channels; both must be closed for operation, and either opening will stop the system, this redundancy ensures safety.
4. Only during emergencies, not for regular stopping of the machine.
5. To ensure it is free from damage or faults and will work properly when needed.

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Answers

Worksheet 3 – Status LED

Questions:

1. What is the purpose of the Status LED in the system?
2. What does a green Status LED indicate?
3. What colour does the LED turn when a fault is detected, but the system may still function?
4. How can you trigger a safety fault to observe the LED behaviour?
5. Why is it important to monitor the Status LED regularly?

Answers:

1. To indicate the system's condition and help quickly identify faults and safety issues for efficient maintenance.
2. Green means the system is operating normally.
3. Orange, it signals a fault that needs investigation.
4. By pressing the Emergency Stop (E-Stop) to put the system into a safety error mode.
5. Because it helps prevent downtime and ensures faults are addressed before escalating.

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Answers

Worksheet 4 – PLC

Questions:

1. What is a PLC and what is it designed for?
2. What do the status LEDs on the PLC indicate?
3. What does it mean if an input LED expected to be on is not lit?
4. What should you observe on the output LEDs when the system starts?
5. How can you tell the flow sensor is working by looking at the PLC?

Answers:

1. A Programmable Logic Controller (PLC) is an industrial computer designed for real-time control of machines and processes in harsh environments.
2. They show real-time feedback of input and output signals, helping verify if components are functioning correctly.
3. It could indicate a faulty sensor, wiring problem, or failed connection.
4. One of the output LEDs will flash briefly, this represents the system's status LED.
5. One of the input LEDs will flash rapidly; if flow is reduced, it slows down, and if flow increases, it nearly goes solid, showing the sensor is active.

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Answers

Worksheet 5 – HMI

Questions:

1. What is the purpose of a Human-Machine Interface (HMI)?
2. Where can you view system faults and alarms on the HMI?
3. What happens when you adjust the High or Low Temperature Cutout settings beyond the current temperature?
4. How does the HMI assist with troubleshooting and maintenance?
5. What real-time information can be seen on the main page of the HMI during system operation?

Answers:

1. To allow operators to interact with the system, view live data, control settings, and diagnose faults.
2. Faults are shown in the Faults section; alarms are shown in the Alarms section of the HMI.
3. The system shuts down and displays a fault; it will not restart until the settings are returned to safe values.
4. It logs alarms, shows sensor readings, lets engineers adjust settings, and helps detect issues like hardware faults or inefficiencies.
5. The main screen displays the current water temperature in the top left, along with the system diagram. It also shows flow rate, valve position, and pump demand in real time.

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Answers

Worksheet 6 – Pump

Questions:

1. What is the function of a pump in an industrial system?
2. How can a pump be controlled to vary its speed?
3. What might it indicate if a pump is running at full speed but still not meeting the setpoint?
4. Why must pumps not be run dry, and what protects against this?
5. What are some signs that maintenance engineers should look for to detect pump issues early?

Answers:

1. To move fluids through a system by creating flow and pressure, supporting process control and system regulation.
2. By varying the voltage (0–24V), using PWM signals, or through a Variable Frequency Drive (VFD) to adjust speed proportionally.
3. It could signal a system fault such as a partially closed valve, clogged filter, or restriction in the piping.
4. Pumps use the fluid as coolant and lubricant; running dry can cause overheating or damage. A low-level sensor in the tank prevents this.
5. Gradual drops in flow with rising power use, unusual noises, or excessive vibration are signs of potential pump problems.

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Answers

Worksheet 7 – Valve

Questions:

1. What is the main function of a valve in an industrial system?
2. How is the valve position controlled in this system?
3. What is meant by the term "deadband" in valve control?
4. What should you do to check if a valve is active during operation?
5. Why is it important to observe valve behaviour during an Emergency Stop?

Answers:

1. To control the flow of fluids by opening, closing, or partially restricting the passage through the system.
2. By using Pulse Width Modulation (PWM) to vary voltage, which proportionally adjusts how open or closed the valve is.
3. Deadband refers to the range (typically 0–10% demand) where the valve remains shut and doesn't begin opening until demand exceeds this threshold.
4. Carefully feel for warmth on the valve's electrical connector , it typically gets warm when active (never check for leaks by hand).
5. To ensure the valve fails to the correct position (open or closed) for safety , in this system, it should go fully closed.

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Answers

Worksheet 8 – Float Switch

Questions:

1. What is the main purpose of a float switch in an industrial system?
2. What is the difference between a normally open (NO) and a normally closed (NC) float switch?
3. Why is the low-level float switch normally open (NO)?
4. Why is the high-level float switch normally closed (NC)?
5. What are some common float switch faults, and how can they be identified?

Answers:

1. To detect liquid levels in a tank and trigger actions such as stopping a pump or sounding an alarm.
2. A NO switch only sends a signal when activated (float is lifted). An NC switch sends a signal when inactivated and breaks the signal when lifted.
3. So, the pump only runs when there is enough fluid. If the level drops, the circuit opens and stops the pump to prevent dry running.
4. So, the system constantly monitors for overflow. If the tank gets too full, the circuit breaks and stops the filling process or triggers an alarm.
5. Sticking, misalignment, or contact failure. These can be identified by testing float movement and checking IO screen signals.

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Answers

Worksheet 9 – Proximity Switch

Questions:

1. What is the main function of a proximity switch in an automated system?
2. What type of object does an inductive proximity sensor detect?
3. Why is the proximity sensor used to detect the tank lid in this system?
4. What are three types of proximity sensors and what do they detect?
5. What common issues can cause a proximity sensor to give false readings?

Answers:

1. To detect the presence of an object without physical contact, helping with automation, positioning, and safety.
2. It detects metal objects only.
3. To ensure the tank lid is in place before the system runs, preventing water from being pumped without the lid.
4. Inductive (metal), capacitive (metal and non-metal), and photoelectric (uses light to detect objects).
5. Misalignment, poor mounting, wiring faults, dust, or moisture.

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Answers

Worksheet 10 – Flow Sensor

Questions:

1. What does a flow sensor measure and why is it important in automated systems?
2. How does a turbine flow sensor work?
3. What role does the PLC play in reading the flow sensor data?
4. What might unstable or incorrect flow readings indicate?
5. Why is regular calibration of flow sensors important?

Answers:

1. It measures fluid movement and helps regulate flow, which is critical for system reliability, detecting faults, and maintaining performance.
2. It uses a spinning rotor, and the number of spins (pulses) is counted to determine the flow rate.
3. The PLC counts pulses from the turbine sensor and calculates the flow rate, which is shown on the HMI.
4. They could indicate blockages, valve restrictions, air pockets, sensor faults, or wiring issues.
5. Calibration ensures the sensor remains accurate over time and helps prevent false readings or system faults.

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Answers

Worksheet 11 – Temperature Sensor

Questions:

1. What is the main purpose of a temperature sensor in an automated system?
2. How does an RTD temperature sensor work?
3. What happens if the temperature exceeds the high cutout limit set on the HMI?
4. What could cause a temperature sensor to give incorrect or no readings?
5. Why is calibration important for temperature sensors?

Answers:

1. To monitor heat levels and prevent overheating or freezing, ensuring safe and efficient operation.
2. It changes resistance based on temperature, which is then converted into a signal the PLC can read.
3. The system will shut down to protect against overheating or unsafe conditions.
4. Possible causes include wiring faults, sensor damage, dirt, vibration, or electrical interference.
5. Calibration keeps the readings accurate and ensures the sensor responds correctly to real conditions.

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Answers

Worksheet 12 – Digital Sensors

Questions:

1. What type of signal does a digital sensor send to a PLC?
2. How can you monitor the status of a digital input?
3. What is the difference between a Normally Open (NO) and Normally Closed (NC) sensor?
4. Why are Emergency Stop buttons typically wired as Normally Closed (NC)?
5. Can you name the seven digital inputs used in the training system?

Answers:

1. A digital sensor sends a simple on/off signal, usually using 24V DC.
2. You can view the input status directly on the PLC or on the IO screen of the HMI.
3. A NO sensor sends a signal only when triggered, while an NC sensor sends a signal when idle and stops when triggered.
4. So that if a wire breaks or becomes disconnected, the system sees it as a fault and shuts down for safety.
5. Flow sensor, tank high switch, tank low switch, tank lid proximity switch, two E-Stop channels, and the blue reset switch.

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Answers

Worksheet 13 – Analogue Sensors

Questions

1. What is the only analogue sensor used on this training rig, and what does it measure?
2. What voltage signal does the temperature sensor send to the PLC, and what temperature range does it represent?
3. What does the PLC do with the analogue voltage signal it receives from the sensor?
4. What happens if the sensor range changes from 0–100°C to 0–300°C but the PLC scaling is not updated?
5. Why might an offset be used, and how does it affect the final temperature value?

Answers

1. The temperature sensor, which measures from 0 to 100°C.
2. It sends a 0–10V signal, where 0V = 0°C and 10V = 100°C.
3. The PLC scales the voltage into a real temperature value using a formula, so it can be shown correctly on the HMI.
4. The temperature readings will be wrong. For example, it may show 100°C when the real temperature is actually 300°C.
5. An offset corrects small errors in readings. For example, if the sensor reads 5°C too high, applying an offset of -5 adjusts it back to the correct value.

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Answers

Worksheet 14 – Faults

Questions

1. Why is fault finding important in industrial maintenance?
2. What are the three main types of faults in industrial systems?
3. How does the HMI help with diagnosing faults?
4. What is the purpose of checking the alarm raise time on the HMI?
5. How can practising with simulated faults improve troubleshooting skills?

Answers

1. It helps identify and fix issues that could stop the system from running safely and efficiently.
2. Software faults (programming or settings), electrical faults (wiring or sensor issues), and mechanical faults (stuck or broken parts).
3. The HMI shows alarms and fault status, helping to locate and understand the issue.
4. It shows exactly when the fault happened, helping to trace the cause.
5. It builds confidence and teaches how to spot and respond to problems in a real system.

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Answers

Worksheet 15 – Lock Out Tag Out

Questions

1. What is the purpose of Lockout Tagout (LOTO)?
2. Why is locking off a valve important before removing downstream pipework?
3. Why should you not rely on PLC logic or software alone for safe isolation?
4. Who should keep the key to the padlock during a LOTO procedure?
5. What should you verify before removing a component downstream of a locked valve?

Answers

1. To ensure equipment is safely isolated so maintenance can be carried out without unexpected operation.
2. It prevents flow through the system, making it safe to remove pipes or components downstream.
3. Software can change or fail, but physical isolation guarantees the system cannot operate.
4. The person carrying out the work should keep the key.
5. That flow has stopped and the system is fully isolated before starting work.

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VERSION CONTROL

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